Redox Reaction

① Hydrogen Economy → Use of liquid hydrogen as fuel.
② Ozone Hole → Depletion of Ozone layer due to CFC's.

Classical Idea of Redox Reaction

Oxidation	Reduction
Addition of Oxygen	Removal of Oxygen
Removal of Hydrogen	Addition of Hydrogen
Addition of more electronegative atom	Removal of electro-negative atom
Removal of electro-positive element	Addition of electro-positive element
$Mg + O_2 \rightarrow MgO$	$HgO \xrightarrow{\Delta} Hg(l) + O_2(g)$
$S + O_2 \rightarrow SO_2$	$FeCl_3 + H_2 \rightarrow FeCl_2 + HCl$
$CH_4 + O_2 \rightarrow CO_2 + H_2O$	$CH_2 = CH_2 + H_2 \xrightarrow{Ni} CH_3 - CH_3$
$H_2S + O_2 \rightarrow S + H_2O$	$HgCl_2 + SnCl_2 \rightarrow \underbrace{Hg_2Cl_2}_{red^n} + \underbrace{SnCl_4}_{oxd^n}$ Redox react^n

Electronic Concept

Trick → OIL — Oxidation is loss of e^- | RIG — Reduction is gain of e^-

$$2Na^0 + Cl_2^0 \longrightarrow 2Na^{+1}Cl^{-1}$$

Gain of e^- (2); loss of e^- (2)

$$2Na^0 + O_2(g) \longrightarrow \overset{+1}{Na_2}O\,(s)$$

loss of $2e^-$; gain of $2e^-$

Half - Reaction

OH $\quad 2Na^0(s) \longrightarrow 2Na^{+1}(g) + 2e^-$ Oxd^n

RH $\quad 2e^- + Cl_2^0(g) \longrightarrow 2Cl^-(g)$ Red^n

Trick →

OP	RR
Oxidation product (loss of e^-)	Reduction reactant (gain of e^-)
Product ke side me	Reactant ke side me

Oxidizing Agent (oxidant) → Oxidizes others and reduces itself (gain of e^-)

$$Cu + H^+ + NO_3^- \rightarrow Cu^{+2} + H_2O + NO_2$$

Reduces itself → Oxidizing agent

Reducing Agent (reductant) → Reduces others and Oxidizes itself (loss of e^-)

$$Sn^{+2} + Fe^{+3} \longrightarrow Sn^{+4} + Fe^{+2}$$

Reduce → Ox^n agent

Oxidize → Reducing agent

Oxidation Number →

Oxidation State →

→ Charge present on atom in a molecule

Rules for Oxidation Number →

Oxidation number of element in elemental / free state is zero.

H_2, O_2, N_2, Fe, Cu, P_4, S_8, Na, Mg, Al

Alkali Metal	Alkaline Earth Metal
G-1 → +1	G-2 → +2
Li, Na, K, Rb, Cs, Fr	Be, Mg, Ca, Sr, Ba, Ra

Ions → Oxidation number = charge

$Na^+ = 1$, $Mg^{+2} = 2$, $O^{2-} = -2$, $F^- = -1$

Oxygen → General oxd^n no. = −2

Peroxide → O^{-1} → H_2O_2, Na_2O_2

Superoxide → $O^{-1/2}$ → KO_2, RbO_2, CsO_2

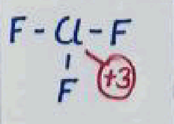
Oxyflouride → +1, +2 → OF_2, O_2F_2

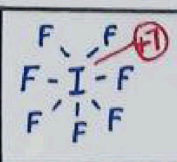
Hydrogen → Generally, it shows +1, in metal hydrides −1

LiH, NaH, CaH_2, MgH_2

Halogen → Generally = −1 F, Cl, Br, I −1

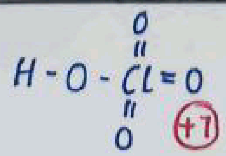
But, Cl, Br, I → +ve Oxidation state (If Bonded with more E.N.)

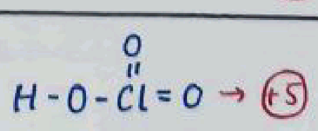




H-O-Cl → +1

H-O-Cl=O → +3





Radical	Name	Oxd^n No
X^-	Halide	-1
O^{-2}	Oxide	-2
N^{-3}	Nitride	-3
P^{-3}	Phosphide	-3
S^{-2}	Sulphide	-2
OH^-	Hydroxide	-1
CN^-	Cyanide	-1
NC^-	Isocyanide	-1
SO_3^{-2}	Sulphite	-2
SO_4^{-2}	Sulphate	-2
PO_3^{-3}	Phosphite	-3
PO_4^{-3}	Phosphate	-3
NO_2^{-1}	Nitrite	-1
NO_3^-	Nitrate	-1
$C_2O_4^{-2}$	Oxalate	-2
$S_2O_3^{-2}$	Thiosulphate	-2
$S_4O_6^{-2}$	Tetrathionate	-2
ClO^-	Hypochlorite	-1
ClO_2^-	Chlorite	-1
ClO_3^-	Chlorate	-1
ClO_4^-	Perchlorate	-1
MnO_4^{-2}	Mangate	-2
MnO_4^-	Permagnate	-1
CrO_4^{-2}	Chromate	-2
$Cr_2O_7^{-2}$	Dichromate	-2
SCN^-	Thiocyanate	-1
NCS^-	Isothiocyanate	-1
NH_4^+	Ammonium	+1
NO^+	Nitrosylium	+1

→ Sum of oxidation number of all elements present in a molecule is equal to the charge on molecule.

e.g. $SO_4^{-2} \rightarrow \cancel{+\theta}\ (ON)_S + 4(ON)_O = -2$

$NH_4^{+2} \rightarrow (ON)_N + 4(ON)_H = +1$

$H_2O \rightarrow 2(ON)_H + (ON)_O = 0$

$Cr_2O_7^{-2} \rightarrow 2x + (-14) = -2 \rightarrow x = 6$

$NH_4^+ \rightarrow x + 4 = 1 \rightarrow x = -3$

$[Fe(CN)_6]^{-3} \rightarrow (-3) - (-6) = x = +3$

$Ni(CO)_5 \rightarrow x + 5 \times 0 = 0 \rightarrow x = 0$

$Fe_3O_4 \rightarrow 3x - 8 = 0 \rightarrow x = 8/3$ → Fractional ON = avg ON

$[Ag(S_2O_3)_2]^{-3} \rightarrow (-3) - (-4) \rightarrow -1$

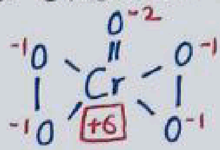
$(NH_4)(NO_3) \rightarrow (NH_4)^+ = x + 4 = 1 \quad x = -3$

$(NO_3)^- = x - 6 = -1 \quad x = 5$

$Fe_3O_4 \nearrow FeO^{+2}$, $\searrow Fe_2O_3^{+3}$ $\quad \dfrac{1 \times (+2) + 2 \times (+3)}{3} = \dfrac{8}{3}$

Exceptional Oxidation Numbers

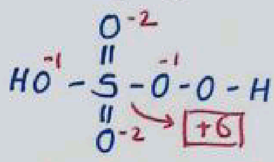
→ CrO_5



Deep blue Butterfly Shape

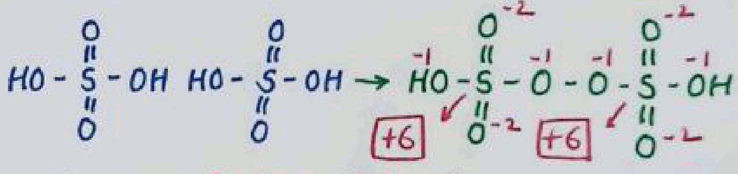
→ $H_2SO_5 \rightarrow H_2SO_4 + O$

Caro's Acid



→ $H_2S_2O_8$ → Marshell's Acid

$H_2SO_4 \times 2 \xrightarrow{-H_2} H_2S_2O_8$

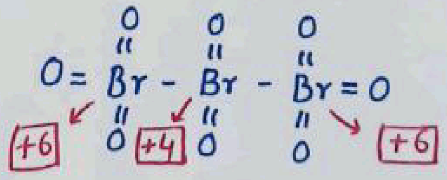


→ C_3O_2 → Carbon Suboxide

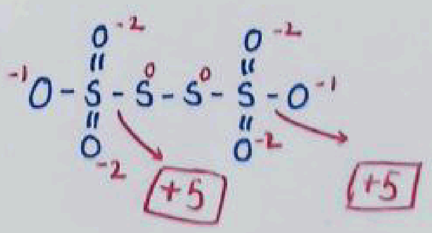
$O = C = C = C = O$

$-2 \quad +2 \quad 0 \quad +2 \quad -2$

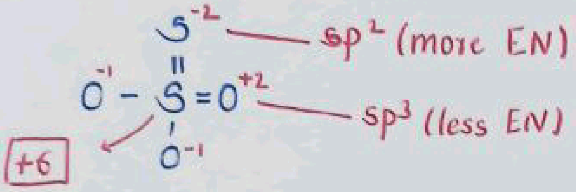
→ Br_3O_8 → Tribromo-octaoxide



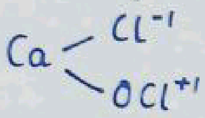
→ $S_4O_6^{-2}$ → Tetrathionate



→ $S_2O_3^{-2}$ → Thiosulphate ion (V. imp)



→ Bleaching Powder → $CaOCl_2$



→ Brown Ring Complex →
$[\overset{+1}{Fe}(H_2O)\overset{+1}{N}\overset{+2}{O}]SO_4^{-2}$

Types of Redox Reaction

Combination Redox Reaction	Decomposition Redox Reaction
2 atoms / molecules combine together to form a product	Reverse of Combination
Atleast 1 reactant must be in elemental form	One product should be in elemental form
$C + O_2 \rightarrow CO_2$	$H_2O \rightarrow H_2 + O_2$
$Mg + N_2 \rightarrow Mg_3N_2$	$NaH \rightarrow Na + H_2$
$CH_4 + O_2 \rightarrow CO_2 + H_2O$	$KClO_3 \rightarrow KCl + 3O_2$
$H_2 + O_2 \rightarrow H_2O$	$LiNO_3 \rightarrow Li_2O + NO_2 + O_2$

Displacement Reaction →

Atom / ion is displayed by another metal atom / ion.

$$AB + C \longrightarrow AC + B$$

Metal displacement	Non-metal displacement
Metal Atom / ion is displaced by another metal	Non metal ion / atom is displaced by another metal / non-metal.
$CuSO_4 + Zn \rightarrow Cu + ZnSO_4$	$Na + H_2O \rightarrow NaOH + H_2$
$Cr_2O_3 + Al \rightarrow Cr + Al_2O_3$	$Ca + H_2O \rightarrow Ca(OH)_2 + H_2$

Disproportionation Reaction →

Rx^n in which oxidation and reduction of same element occurs from oxidation number.

$$\underbrace{\underbrace{H_2\overset{-1}{O_2} \longrightarrow H_2\overset{-2}{O}}_{Red^n} + \overset{0}{O_2}}_{Ox^n}$$

$$\underbrace{\underbrace{\overset{0}{P_4} + NaOH \rightarrow \overset{-3}{P}H_3}_{Red^n} + NaH_2\overset{+1}{P}O_2}_{Oxd^n}$$

$Cl_2 + NaOH \longrightarrow Na\overset{-1}{Cl} + NaO\overset{+1}{Cl}$
(cold & dilute)

$Cl_2 + NaOH \longrightarrow Na\overset{-1}{Cl} + Na\overset{+5}{Cl}O_3$
(hot & conc.)

Hypochlorite	$HClO \rightarrow HCl + HClO_2$
Chlorite	$HClO_2 \rightarrow HCl + HClO_3$
Chlorate	$HClO_3 \rightarrow HCl + HClO_4$
Perchlorate	$HClO_4 \rightarrow X$

Comproportionation Reaction →

Reverse of Dispropⁿ Reaction

$\overset{-3}{N}H_4\overset{+3}{N}O_2 \rightarrow \overset{0}{N_2} + H_2O$

Balancing of Redox Reaction

→ atom, charge balance

Ion e⁻ method	Oxdⁿ no. method
1st → atom balance	1st → charge balance
2nd → charge balance	2nd → atom balance

Ion e⁻ Method →

eg → Balance Rx^n in Basic Method

$\overset{0}{Zn} + NO_3 \rightarrow \overset{+2}{Zn} + NH_4^+$

Step-1 → Breakup (Oxyⁿ Half & Redⁿ Half)

$Zn \rightarrow Zn^{+2}$ (O.H)

$\overset{+5}{N}O_3^- \rightarrow \overset{-3}{N}H_4^+$ (R.H)

Step-2 → Balance atom other than O & H

$Zn \rightarrow Zn^{+2}$

$\overset{+5}{N}O_3^- \rightarrow \overset{-3}{N}H_4^+$

Step-3 → Balance O → H_2O, H → H^+

$\overset{0}{Zn} \rightarrow Zn^{+2}$

$10H^+ + NO_3^- \rightarrow NH_4^+ + 3H_2O$

Step-4 → (Basic medium) Add OH^-

$Zn \rightarrow Zn^{+2}$

$10H^+ + NO_3^- \longrightarrow NH_4^+ + 3H_2O + 10\,OH^-$

$10H_2O + NO_3^- \longrightarrow NH_4^+ + 3H_2O + 10\,OH^-$

$7H_2O + NO_3^- \longrightarrow NH_4^+ + 10\,OH^-$

$Na_2S_2O_3$ (Reducing agent)

$Na_2S_2O_3 \rightarrow Na_2S_4O_6$

$nf = 0.5 \times 2 = 1$ | $nf = 0.5 \times 4 = 2$

$\overset{0}{I_2} \rightarrow \overset{+5}{IO_3^-}$ = $nf = 5 \times 2 = 10$

$\overset{-2}{N_2H_4} \rightarrow \overset{+3}{HNO_2}$ = $nf = 5 \times 2 = 10$

$\overset{+7}{IO_4^-} \rightarrow \overset{0}{I_2}$ = $nf = 7 \times 1 = 7$

$\overset{+2}{VO} \rightarrow \overset{+5}{V_2O_5}$ = $nf = 3 \times 1 = 3$

$\overset{+3}{Fe}(\overset{+3}{C_2O_4})_3 \rightarrow Fe^{+3} + \overset{+4}{CO_2}$ = $nf = 6 \times 1 = 6$

$Na_2\overset{+2}{S_2}O_3 + I_2 \rightarrow Na_2S_4O_6 + 2NaI$ =

$nf = 0.5 \times 2 = 1$ | $nf = 1 \times 2 = 2$

When more than 1 O.S. is changing →

Case-1 → If Oxd^n no. either ↑ or ↓

$\overset{+2}{Fe}\,\overset{+3}{C_2O_4} \rightarrow Fe^{+3} + \overset{+4}{CO_2}$

2 ↑ 2 ↑

$nf = |n_1 + n_2| = 2 + 2 = 4$

$\overset{+2}{Fe}\overset{-1}{S_2} \rightarrow \overset{+3}{Fe_2O_3} + \overset{+4}{SO_2}$ $nf = 11$

1 10

$\overset{+1}{Cu_2}\overset{-2}{S} \rightarrow \overset{+2}{Cu} + \overset{+4}{SO_2}$ $nf = 8$

2 6

Case-2 → If Oxd^n no. ↑ and ↓

$\overset{0}{Br_2} \rightarrow \overset{-1}{Br^-} + \overset{+5}{BrO_3^-}$

10↑ 2↓

$nf = 8$

disprop → $nf = \frac{n_1 \cdot n_2}{n_1 + n_2} = \frac{20}{12} = \frac{5}{3}$

$\overset{0}{P_4} \rightarrow \overset{-3}{PH_3} + NaH_2\overset{+1}{PO_2}$

12 4

$nf = 8$

$nf = 3$ $nf = 1$

disprop

$nf = \frac{48}{16}$, $nf = 3$

Equivalent Mass

$$\boxed{Eq^t\ Mass = \frac{Molar\ Mass}{n\text{-}factor}}$$

eg. —

$\overset{0}{Mg} + \frac{1}{2}\overset{0}{O_2} \rightarrow \overset{+2}{Mg}\overset{-2}{O}$

$nf = 2$

$Eq^t\ mass = \frac{24}{2} = 12$

$nf = 4$

$Eq^t\ mass = \frac{32}{4} = 8$

no. of mole = $\frac{GM}{mm}$	No. of eq^t = $\frac{GM}{eq^t\ mass}$ = $\frac{GM}{MM} \times n.factor$ No. of Eq^t = mole × n. factor
Molarity → = $\frac{\text{no. of moles of solute}}{\text{vol. of sol}^n \text{ in litre}}$	Normality → = $\frac{\text{no. of eq}^t \text{ of solute}}{\text{vol. of sol}^n \text{ (litre)}}$
No. of mole → = M × Vol (lit)	No. of eq^t → = N × Vol (lit)

Step 5 → Charge Balance

$(Zn^0 \rightarrow Zn^{+2} + 2e^-) \times 4$

$8e^- + 7H_2O + \overset{+5}{N}O_3^- \rightarrow \overset{-3}{N}H_4^+ + 10OH^-$

$4Zn \rightarrow 4Zn^{+2} + \cancel{8e^-}$

$\cancel{8e^-} + 7H_2O + \overset{5+}{N}O_3^- \rightarrow \overset{-3}{N}H_4^+ + 10OH^-$

$4Zn + 7H_2O + NO_3^- \rightarrow 4Zn^{+2} + NH_4^+ + 10OH^-$

Ans ↑

Balance Reaction in Acidic Medium

$Fe^{+2} + Cr_2O_7^{-2} \rightarrow Fe^{3+} + Cr^{+3}$

Step 1 → $Fe^{+2} \rightarrow Fe^{3+}$ (O.H)

$\overset{+6}{Cr_2}\overset{-2}{O_7}^{-2} \rightarrow Cr^{+3}$ (R.H)

Step 2 → $Fe^{+2} \rightarrow Fe^{3+}$

$\overset{+6}{Cr_2}\overset{-2}{O_7} \rightarrow 2Cr^{+3}$

Step 3 → $Fe^{+2} \rightarrow Fe^{+3}$ Oxy^n

$14H^+ + \overset{+6}{Cr_2}\overset{-2}{O_7} \rightarrow 2Cr^{+3} + 7H_2O$ Red^n

Step 4 → $(Fe^{+2} \rightarrow Fe^{+3} + e^-) \times 6$

$+6e^- + 14H^+ + \overset{+6}{Cr_2}\overset{-2}{O_7} \rightarrow 2Cr^{+3} + 7H_2O$

$6Fe^{+2} \rightarrow 6Fe^{+3} + 6e^-$

$+6e^- + 14H^+ + \overset{+6}{Cr_2}\overset{-2}{O_7} \rightarrow 2Cr^{+3} + 7H_2O$

$6Fe^{+2} + 14H^+ + \overset{+6}{Cr_2}\overset{-2}{O_7} \rightarrow 6Fe^{+3} + 2Cr^{+3} + 7H_2O$

Ans ↑

Oxidation Number Method

$\overset{+7}{Mn}O_4^- + C_2O_4^{-2} + H^+ \rightarrow CO_2 + Mn^{2+} + H_2O$

5 2 (Reverse)

$2MnO_4^- + 5C_2O_4^{-2} + H^+ \rightarrow 10CO_2 + 2Mn^{2+} + H_2O$

$2MnO_4^- + 5C_2O_4^{-2} + 16H^+ \rightarrow 10CO_2 + 2Mn^{2+} + 8H_2O$

n-factor / Valence factor / x-factor

Calculation →

1 Ions → n-factor = |charge|.

Na^+, Mg^{+2}, Al^{+3}, SO_4^{-2}

1 2 3 4

2 Salt → n-factor = LCM of cationic charge & anionic charge

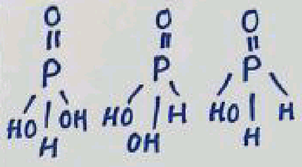
$NaCl \rightarrow$ 1, $MgO \rightarrow$ 2, $\overset{+3}{Al_2}(\overset{-2}{SO_4})_2 \rightarrow$ 6

3 Acid / Base →

Acid	Base
n-factor = basicity → no. of H^+ donated	n-factor = acidity → no. of OH^- donated → no. of H^+ accepted
HCl, HBr, HI, $HClO_4$, CH_3COOH nf = 1	$NaOH$, KOH, $RbOH$, $CsOH$, NH_3, RNH_2, $PhNH_2$ nf = 1
H_2SO_4, H_2CO_3, $H_2C_2O_4$ nf = 2	$Mg(OH)_2$, $Ca(OH)_2$, en, $Zn(OH)_2$ nf = 2
$H_3PO_4 \rightarrow$ nf = 3	$Al(OH)_3$, $Fe(OH)_3$ nf = 2

Phosphorous acid Series →

H_3PO_4	H_3PO_3	H_3PO_2
nf = 3	nf = 2	nf = 1
$3H^+$ donate	$2H^+$ donate	$1H^+$ donate



Boric Acid →

$H_3BO_3 / B(OH)_3 \rightarrow$ nf = 1

(1 Hydrogen donated from H_2O).

Oxidizing Agent and Reducing Agent (n factor)

$KMnO_4 \rightarrow Mn^{+2}$ acidic nf = 5

$KMnO_4 \rightarrow \overset{+6}{Mn}\overset{-2}{O_4}$ basic nf = 1

$KMnO_4 \rightarrow \overset{+4}{Mn}O_2$ neutral nf = 3

$K_2\overset{+6}{Cr_2}O_7 \xrightarrow{\text{acidic}} Cr^{+3}$ nf = 6

$Fe^{+2} \rightarrow Fe^{+3}$

$C_2O_4^{2-} \rightarrow CO_2$

$I^- \rightarrow I_2$

$H_2S \rightarrow S$

$Sn^{2+} \rightarrow Sn^{4+}$

$H_2O_2 \rightarrow O_2$

$SO_2 \rightarrow SO_4^{2-}$

H_2O_2

Oxyd^n agent	Red^n agent
Self red^n	Self oxy^n
$H_2O_2^{-1} \rightarrow H_2O^{-2}$	$H_2O_2^{-1} \rightarrow O_2^0$
nf = 1 × 2 = 2	nf = 2 × 1 = 2